Synthetic Prime-Trace Operators in the PEN Universe: Rigged Traces and the Critical-Line Spectral Side

Halvor Lande

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Abstract

We work inside the fully coupled univalent PEN universe at steps 14 and 15, where the selected library already contains inner products, orthogonal decomposition, spectral calculus for self-adjoint operators, metric/Hodge structure, cohesion, and the temporal-cohesive shell. In that environment we construct a canonical prime-trace operator directly from geometry. The carrier is the synthetic idele-class space

$$X_\zeta := \mathbb{A}^\times/\mathbb{Q}^\times,$$

assembled top-down from the geometric circle, its loop group, Stone completion, the derived real line, and quotient formation already available in the library. The step-14 Hilbert package yields the critical Hilbert space

$$V_\zeta \subseteq L^2\Omega^*(X_\zeta, g_\zeta),$$

while the step-10 cohesive shell supplies a codiscrete nuclear test module

$$\Phi_\zeta \hookrightarrow V_\zeta$$

with dense inclusion into that Hilbert carrier. Here V_ζ is not specified by its intended spectral role: it is obtained by taking the Hilbert closure of the exact kernel intersection of the step-13 volume map and the step-15 archimedean jet map on the full Schwartz carrier. The step-15 temporal shell internalizes the scaling flow of $\mathbb{R}_{>0}$ as a strongly continuous unitary action. Its Stone generator

$$H_\zeta = -i\mathcal{L}_{v_{\mathrm{scale}}}\big|_{V_\zeta}$$

is self-adjoint.

The main trace theorem is stated in regularized form:

$$\mathrm{Tr}_{\mathrm{reg}}(U_\Phi(h)) = W(h)$$

for every compactly supported smooth multiplicative test function h, where $U_\Phi(h)$ is the integrated scaling action on the test module, $\mathrm{Tr}_{\mathrm{reg}}$ is the categorical even-minus-odd trace exported by the graded step-10/14/15 interface, and W is the standard Weil distribution. The trace is not evaluated on the raw Hilbert carrier: the noncompact L^2 object is used only for self-adjoint spectral calculus, while the explicit formula is realized on the dualizable nuclear side of the rigged pair $\Phi_\zeta \hookrightarrow V_\zeta$. Thus no claim is made that an orthogonal compression of $L^2(X_\zeta)$ is trace class or that a classical Lefschetz theorem applies directly to the noncompact quotient. Since the generator is self-adjoint, its spectral parameter is real, and the realized spectral side is supported on $1/2 + i\mathbb{R}$. This is a representation theorem for the rigged trace datum, not a proof of the converse direction of Weil's criterion: no claim is made here that the full classical zero distribution has been recovered from the conditionally convergent zero sum. We then show that the same operator is also the unique PEN-optimal realization of the prime-trace API: any off-line branch requires at least three extra MBTT clause families—pairing, reflection, and non-self-adjoint resonance calculus—and contributes no additional public novelty. Thus the Hermitian critical-line spectral package is both the geometric consequence of the rigged prime-trace realization and the unique structurally admissible operator branch in the PEN universe.

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1 Introduction

The Riemann Hypothesis asserts that every non-trivial zero of the Riemann zeta function has real part 1/2. In the classical literature, the strongest geometric route toward this statement is the Hilbert–Pólya idea: if one can exhibit a self-adjoint operator whose spectral side reproduces the explicit formula, the critical line follows from self-adjointness. Connes' adelic trace-formula program reframed this in geometric terms: the prime distribution is encoded by the scaling action on adele classes, and the explicit formulas appear as a Lefschetz trace formula on the resulting noncommutative space [Connes, 1999, Connes et al., 2009]. Meyer subsequently gave a cohomological spectral realization of the zeros on a nuclear Fréchet space [Meyer, 2005]. What remains difficult in the classical picture is not the formal explicit formula itself, but the attempt to force that formula through an ordinary trace-class calculus on a raw Hilbert space.

The present paper does not claim to solve that analytic promotion inside classical ZFC. Instead, it changes the typing of the problem. The trace side lives on a step-10 codiscrete/nuclear test module Φ_ζ, while the spectral side lives on its step-14 Hilbert completion V_ζ. The bridge between them is a PEN-internal rigged Hilbert datum

$$\Phi_\zeta \hookrightarrow V_\zeta \hookrightarrow \Phi_\zeta^\vee,$$

not a singular-value estimate for a spatially compressed operator on a noncompact quotient.

The PEN universe supplies exactly the missing infrastructure. By the earlier PEN papers, framework evolution in the fully coupled univalent lane reaches, at step 14, a Hilbert-functional package with inner products, completeness, orthogonal decomposition, spectral calculus, and operator bridges; at step 15 it reaches a temporal-cohesive shell whose next modality internalizes infinitesimal evolution as a tangent-style endofunctor [Lande, 2026c,e,b,d]. The late library therefore contains the objects needed to state and solve the Hilbert–Pólya problem synthetically.

The theorem proved in this paper is the following.

Theorem 1.1 (Main theorem, informal)**.** *Inside the step-14/15 PEN universe there exists a canonical rigged prime-trace datum*

$$\Phi_\zeta \hookrightarrow V_\zeta$$

preserved by scaling, with V_ζ a graded Hilbert object and Φ_ζ a dualizable nuclear test module. The Stone generator H_ζ on V_ζ is self-adjoint, the categorical supertrace of the induced action on Φ_ζ realizes the Weil explicit distribution, and the same regularized trace admits a spectral representation supported on $1/2 + i\mathbb{R}$. This separates the two classical tasks: exact trace evaluation is carried out on the dualizable test side, while spectral reality is proved on the Hilbert side. A further converse argument of Weil-criterion type would be needed to identify that spectral measure with the full classical zero distribution, and that step is not claimed here.

The argument has two logically separate parts.

First, we construct the rigged prime-trace datum and prove the regularized trace formula. The synthetic idele-class carrier is built from the geometric core of the selected PEN trajectory: the circle provides $\mathbb{Z} \simeq \Omega S^1$, Stone completion provides the profinite arithmetic boundary, the real-cohesive prelude provides the archimedean line, the step-10 cohesive shell provides a codiscrete Schwartz-type module, and the step-13/14 metric-Hilbert package upgrades the same quotient to a graded Hilbert object. The critical summand is then defined explicitly as the intersection of the kernel of the step-13 volume functional with the kernel of the step-15 archimedean jet evaluator, and V_ζ is its Hilbert closure. The temporal shell turns scaling into a strongly continuous unitary action, whose Stone generator is self-adjoint on V_ζ. The corresponding explicit formula is evaluated as a categorical supertrace on Φ_ζ, not as a Fredholm trace of a compressed L^2 operator.

Second, we analyze the same operator through the MBTT lens used by the PEN synthesis engine. Relative to a fixed public prime-trace API, the Hermitian shell costs exactly two branch-specific clauses, whereas any off-line realization requires at least three additional repair-clause families. These extra clauses add no public novelty, so the PEN objective strictly selects the Hermitian branch.

This makes the critical-line spectral ansatz appear twice: geometrically through self-adjoint spectral realization, and combinatorially through the minimality of the Hermitian operator AST.

1.1 Background imported from the PEN sequence

We use the previously established PEN trajectory as background rather than re-proving it here. The imported facts are:

1. the coherence depth is $d = 2$, hence interface debt obeys the Fibonacci recurrence;
2. the first nine selected layers are the geometric core through the Hopf fibration;
3. steps 10–14 yield cohesion, connections, curvature, metric structure, and the Hilbert-functional package;
4. step 15 yields the temporal-cohesive shell with exchange laws and guarded composition.

These inputs are documented in the companion manuscripts and in the public mechanization repository, whose current README describes a two-pillar artifact with Cubical Agda for proof obligations and Haskell for executable synthesis and scoring [Lande, 2026c,e,d,f,a].

1.2 Plan

Section 2 fixes the step-14/15 baseline and the PEN operator API. Section 3 constructs the synthetic idele-class carrier and classifies the closed scaling orbits. Section 4 lifts this geometry into the step-14 Hilbert package and defines the self-adjoint generator H_ζ. Section 5 proves the prime-trace theorem in regularized cohomological form and derives the critical-line spectral representation. Section 6 then shows that the same operator is the unique PEN-optimal shell. Section 7 assembles the rigged synthesis and isolates the remaining converse step.

2 Step-14/15 baseline and the operator API

2.1 The late PEN universe

Assumption 2.1 (Late PEN universe)**.** We work in the fully coupled univalent PEN universe through step 15. Concretely, the library state $\mathcal{B}_{15}$ contains:

1. the geometric core up to the Hopf fibration;
2. the cohesive shell with $\flat$, $\sharp$, Disc, and Π_∞;
3. the differential shells of connection and curvature;
4. the metric and Hilbert-functional shells, providing inner products, completeness, orthogonal decomposition, and spectral calculus for self-adjoint operators;
5. the temporal shell with next/eventually modalities, exchange laws with the cohesive interface, and guarded coiteration.

Remark 2.2. The present manuscript treats these late structures as already selected. Their stagewise optimality and the real-cohesive scalar provenance were established in the companion framework-abstraction paper [Lande, 2026d].

2.2 The public prime-trace API

The operator that links primes to the continuum should not be specified as a raw trace of a fixed endomorphism. The public object is the scaling family produced by functional calculus.

Definition 2.3 (Multiplicative test algebra)**.** Let $\mathscr{S} := C_c^\infty(\mathbb{R}_{>0})$ be the space of smooth compactly supported test functions on the multiplicative half-line. The multiplicative convolution and involution are

$$(f * g)(x) := \int_0^\infty f(y)g(x/y)\, d^\times y, \qquad f^*(x) := \overline{f(x^{-1})}.$$

Its Mellin transform is

$$\widehat{f}(s) := \int_0^\infty f(x)x^s\, d^\times x.$$

Proposition 2.4 (Why the raw Fredholm trace is unavailable)**.** *Suppose an operator H_ζ has discrete spectrum that faithfully parameterizes the ordinates of the non-trivial zeros of $\zeta(s)$. Then H_ζ is not trace class.*

Proof. Let $\rho_n = \beta_n + i\gamma_n$ enumerate the non-trivial zeros by nondecreasing $|\gamma_n|$. The Riemann–von Mangoldt formula gives

$$N(T) = \#\{\rho : 0 < \Im(\rho) \le T\} = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$$

[Titchmarsh, 1986, Bombieri, 2000]. Hence $|\gamma_n|$ grows on the order of $n/\log n$, so the spectral parameter is not summable. A spectrally faithful zeta operator therefore cannot be trace class. □

Definition 2.5 (Prime-trace API)**.** A *prime-trace realization* over $\mathcal{B}_{15}$ consists of:

1. a graded Hilbert object
$$V_\zeta = V_\zeta^{\text{even}} \oplus V_\zeta^{\text{odd}};$$
2. a graded dualizable nuclear cohesive module
$$\Phi_\zeta = \Phi_\zeta^{\text{even}} \oplus \Phi_\zeta^{\text{odd}}$$
together with a grading-preserving dense inclusion
$$\iota : \Phi_\zeta \hookrightarrow V_\zeta;$$
3. a strongly continuous unitary representation
$$U^V : \mathbb{R}_{>0} \to \mathcal{U}(V_\zeta);$$
preserving the grading, together with compatible endomorphisms
$$U_x^\Phi : \Phi_\zeta \to \Phi_\zeta$$
such that $\iota \circ U_x^\Phi = U_x^V \circ \iota$ for all $x > 0$;
4. its Stone generator H_ζ on V_ζ, so that $U_x^V = x^{iH_\zeta}$;
5. for every $h \in \mathscr{S}$, integrated endomorphisms
$$U_V(h) := \int_0^\infty h(x)U_x^V\, d^\times x, \qquad U_\Phi(h) := \int_0^\infty h(x)U_x^\Phi\, d^\times x;$$
together with the regularized supertrace value
$$\mathrm{Tr}_{\text{reg}}(U_\Phi(h)) := \mathrm{Tr}_{\text{cat}}(U_\Phi(h)|_{\Phi_\zeta^{\text{even}}}) - \mathrm{Tr}_{\text{cat}}(U_\Phi(h)|_{\Phi_\zeta^{\text{odd}}});$$

6. the evaluation law

$$\Theta(h) : \mathrm{Tr}_{\mathrm{reg}}(U_\Phi(h)) = W(h),$$

where W is the classical Weil distribution attached to the Riemann zeta function.

Remark 2.6. This API is intentionally rigged rather than compressed. The paper does not claim that an orthogonal cutoff on a raw L^2 space produces a trace-class operator with the correct trace, nor that a classical Lefschetz trace theorem applies directly to the noncompact quotient X_ζ. In classical analysis that would require hard control of the continuous spectrum at spatial infinity, and in the PEN framework such a bottom-up repair falls under the classical analytic surcharge theorem of the companion framework-abstraction paper [Lande, 2026d, Theorem 2.1]. The public trace datum therefore lives on the dualizable test object Φ_ζ, while the Hilbert object V_ζ is reserved for self-adjoint spectral calculus.

3 The synthetic idele-class carrier

The first task is to construct the arithmetic phase space inside the existing late library without paying the classical bottom-up analytic surcharge. The construction is top-down and geometric.

3.1 From the circle to arithmetic carriers

Definition 3.1 (Geometric integers)**.** Set

$$Z_{\mathrm{geom}} := \Omega S^1.$$

By the standard circle computation in homotopy type theory, Z_{geom} is a discrete group equivalent to $\mathbb{Z}$ [Licata and Shulman, 2017].

Lemma 3.2. *There is an equivalence of discrete groups*

$$Z_{\mathrm{geom}} \simeq \mathbb{Z}.$$

Proof. The encode-decode computation for the universal cover of the circle identifies the loop space at the basepoint with the integers. Group structure comes from loop concatenation on ΩS^1 and addition on $\mathbb{Z}$. □

Definition 3.3 (Stone completion)**.** For each $n \geq 1$, let

$$C_n := Z_{\mathrm{geom}}/nZ_{\mathrm{geom}}.$$

Inside the Stone/condensed subuniverse of the cohesive model, define the profinite completion

$$\widehat{Z} := \varprojlim_n C_n.$$

For each prime p, let $\widehat{Z}_p$ be the p-primary factor, Q_p its fraction object, and define the restricted product

$$\mathbb{A}_f := \prod_p{}' Q_p$$

with respect to the compact open subobjects $\widehat{Z}_p$.

Proposition 3.4 (Derived arithmetic carrier)**.** *The objects $\widehat{Z}$, Q_p, and $\mathbb{A}_f$ are definable in the late PEN universe without introducing new selected primitive clauses.*

Proof. Each C_n is a quotient of the already available group object Z_{geom}, and the inverse limit defining $\widehat{Z}$ is formed inside the compact totally disconnected subuniverse validated by synthetic Stone duality [Cherubini et al., 2025]. Likewise, Q_p and the restricted product $\mathbb{A}_f$ are constructed from localizations, products, and subobject conditions over those already-defined carriers. These operations create derived objects in the closure of $\mathcal{B}_{15}$; they are not new promoted shells, and therefore incur no independent PEN selection charge. □

Definition 3.5 (Archimedean line)**.** Let $\mathbb{R}$ be the derived line object supplied by the real-cohesive prelude of the framework-abstraction phase. Set

$$\mathbb{A} := \mathbb{R} \times \mathbb{A}_f.$$

Definition 3.6 (Synthetic idele-class space)**.** The *synthetic idele-class carrier* is

$$X_\zeta := \mathbb{A}^\times / \mathbb{Q}^\times.$$

The multiplicative group $\mathbb{R}_{>0}$ acts by scaling the archimedean coordinate:

$$\lambda \cdot [(a_\infty, a_f)] := [(\lambda a_\infty, a_f)].$$

3.2 Closed orbit classification

Proposition 3.7 (Primitive closed orbits are primes)**.** *The primitive closed orbits of the $\mathbb{R}_{>0}$-action on X_ζ are canonically indexed by prime numbers. The m-fold iterate of the orbit attached to p has period $m \log p$.*

Proof. A point $[a] \in X_\zeta$ lies on a closed orbit of period $T > 0$ exactly when there exists $q \in \mathbb{Q}^\times$ with

$$(e^T a_\infty, a_f) = (q a_\infty, q a_f)$$

inside $\mathbb{A}^\times$. Equivalently,

$$e^T = |q|_\infty$$

and the finite component returns after multiplication by q. Write

$$q = \pm \prod_p p^{m_p}.$$

If more than one m_p is nonzero, the return decomposes as a sum of shorter independent valuation loops, so the corresponding orbit is not primitive. The primitive case is therefore $q = p$ for a unique prime p, with period

$$T = \log |p|_\infty = \log p.$$

Iterating the same orbit m times corresponds to $q = p^m$, hence to the period $m \log p$. □

Remark 3.8. This is the internal PEN-universe version of the periodic-orbit dictionary in Connes' adelic trace formula: prime numbers label primitive closed orbits and prime powers arise as iterates [Connes, 1999, Connes et al., 2009].

4 Hilbert lift and the self-adjoint generator

4.1 The critical graded Hilbert space

Definition 4.1 (Full Hilbert complex)**.** Let g_ζ be the scale-invariant metric supplied by the step-13 metric shell. Define

$$V_{\text{full}} := L^2\Omega^*(X_\zeta, g_\zeta).$$

4.2 The synthetic Schwartz side

Definition 4.2 (Codiscrete Schwartz carrier)**.** Let $\overline{X}_\zeta$ be the archimedean scaling compactification of X_ζ, and let $x_\infty \in \overline{X}_\zeta$ denote its scale-fixed archimedean boundary point. Let $\Omega^\infty(X_\zeta)$ denote the cohesive module of smooth differential forms on X_ζ. The step-10 codiscrete modality defines the full test carrier

$$\Phi_{\text{full}} := \sharp\Omega^\infty(X_\zeta).$$

Its exported Schwartz boundary discipline gives canonical restriction maps from Φ_{full} to the archimedean compactification and its formal neighborhoods.

Definition 4.3 (Pole and jet morphisms)**.** Define the step-13 pole functional

$$I_{\text{vol}} : \Phi_{\text{full}} \to \mathbb{R}, \qquad I_{\text{vol}}(\phi) := \int_{X_\zeta} \phi^\top,$$

where $\phi^\top$ denotes the top-degree component with respect to g_ζ. Let D_∞ be the step-15 formal neighborhood of x_∞ in $\overline{X}_\zeta$, and define the archimedean jet evaluator

$$J_\infty : \Phi_{\text{full}} \to \Omega^\infty(D_\infty), \qquad J_\infty(\phi) := \phi|_{D_\infty}.$$

The first map packages the pole contribution, while the second packages the archimedean jet data responsible for the trivial-zero sector.

Definition 4.4 (Explicit critical carrier)**.** The critical test module is defined by the exact kernel intersection

$$\Phi_\zeta := \ker(I_{\text{vol}}) \cap \ker(J_\infty).$$

Let $V_\zeta \subseteq V_{\text{full}}$ be the Hilbert closure of the image of Φ_ζ under the natural smooth-to-L^2 map, and set

$$V_{\text{triv}} := V_\zeta^\perp.$$

Write P_ζ for the orthogonal projector onto V_ζ. Thus the step-14 orthogonal-decomposition clause is used only after V_ζ has been constructed as a concrete closed subspace.

Proposition 4.5 (Closed critical sector from explicit kernels)**.** *The maps I_{vol} and J_∞ are continuous linear morphisms. Consequently Φ_ζ is a closed graded submodule of Φ_{full}, V_ζ is a closed graded subspace of V_{full}, and*

$$V_{\text{full}} = V_\zeta \oplus V_{\text{triv}}.$$

Moreover V_{triv} is the closed Hilbert sector generated by the concrete pole and jet data encoded by I_{vol} and J_∞; it is not postulated from desired spectral properties.

Proof. Integration against the step-13 volume form is a continuous linear functional on the codiscrete Schwartz carrier, and restriction to a step-15 formal neighborhood is a continuous linear map by functoriality of the cohesive tangent shell. Kernels of continuous linear maps are closed, hence so is their intersection Φ_ζ. By definition V_ζ is the Hilbert closure of that explicit smooth core, so it is closed in V_{full}. The step-14 Hilbert package may therefore be applied non-circularly to this already-constructed closed subspace, producing the orthogonal complement $V_{\text{triv}} = V_\zeta^\perp$ and the splitting

$$V_{\text{full}} = V_\zeta \oplus V_{\text{triv}}.$$

This construction defines the critical carrier from exact kernels first and only then forms its orthogonal complement. □

Definition 4.6 (Parity splitting)**.** The Hodge parity decomposition yields

$$V_\zeta = V_\zeta^{\text{even}} \oplus V_\zeta^{\text{odd}}, \qquad \Phi_\zeta = \Phi_\zeta^{\text{even}} \oplus \Phi_\zeta^{\text{odd}}.$$

The scaling action preserves this grading because it acts by pullback on differential forms and commutes with degree parity.

Proposition 4.7 (Synthetic rigged Hilbert datum)**.** *The inclusion*

$$\iota : \Phi_\zeta \hookrightarrow V_\zeta$$

is dense and grading-preserving. Moreover Φ_ζ is dualizable in the symmetric monoidal category of cohesive modules exported by the late PEN shell, so

$$\Phi_\zeta \hookrightarrow V_\zeta \hookrightarrow \Phi_\zeta^\vee$$

is a PEN-internal rigged Hilbert object.

Proof. The codiscrete modality supplies the full Schwartz-type test carrier, and theorem 4.5 identifies Φ_ζ as a closed graded submodule cut out by explicit continuous boundary maps. The space V_ζ was defined as the Hilbert closure of that critical smooth core, so the inclusion is dense by construction and grading-preserving by the Hodge parity split. Dualizability is the categorical form of nuclearity for this codiscrete test carrier and is part of the step-10/15 cohesive interface used by the prime-trace API. □

Lemma 4.8 (Scaling invariance)**.** *The subspaces V_{triv} and V_ζ, and the dense test module Φ_ζ, are invariant under the scaling action of $\mathbb{R}_{>0}$; hence P_ζ commutes with that action.*

Proof. The scale-invariant metric makes the volume functional homogeneous under pullback by scaling, so $\ker(I_{\text{vol}})$ is invariant. The scaling flow fixes the archimedean boundary point x_∞ and therefore preserves its formal neighborhood D_∞; functoriality gives

$$J_\infty(\phi \circ \text{scale}_{\lambda^{-1}}) = J_\infty(\phi) \circ \text{scale}_{\lambda^{-1}}\big|_{D_\infty}.$$

Hence $\ker(J_\infty)$ is invariant as well, and therefore so is their intersection Φ_ζ. By construction V_ζ is the Hilbert closure of an invariant dense core, so V_ζ is invariant; its orthogonal complement $V_{\text{triv}} = V_\zeta^\perp$ is then invariant because the scaling action is unitary on V_{full}. Since P_ζ is the orthogonal projector onto an invariant closed summand, it commutes with the action. □

4.3 The temporal-cohesive flow

Definition 4.9 (Scaling representation)**.** For $\lambda \in \mathbb{R}_{>0}$ define

$$U_\lambda^V := (\text{scale}_{\lambda^{-1}})^*$$

on V_{full} and hence, by restriction, on V_ζ. Define also

$$U_\lambda^\Phi := U_\lambda^V\big|_{\Phi_\zeta}$$

on the dense test module.

Proposition 4.10 (Unitary action)**.** *The family $(U_\lambda^V)_{\lambda>0}$ is a strongly continuous unitary representation of $\mathbb{R}_{>0}$ on V_ζ.*

Proof. The action law is immediate from pullback functoriality. The scale-invariant metric package provides an L^2-norm on forms that is preserved by the scaling diffeomorphisms, so each U_λ^V is unitary. Strong continuity follows from differentiability of the flow on smooth compactly supported forms and density in the Hilbert completion. □

Theorem 4.11 (Self-adjoint generator)**.** *There exists a unique self-adjoint operator H_ζ on V_ζ such that*

$$U_\lambda^V = \lambda^{iH_\zeta} \qquad (\lambda > 0).$$

On the common core of smooth compactly supported differential forms,

$$H_\zeta = -i\mathcal{L}_{v_{\text{scale}}}\big|_{V_\zeta},$$

where v_{scale} is the infinitesimal scaling vector field.

Proof. By theorem 4.10, the logarithmic reparameterization $t \mapsto U_{e^t}^V$ is a strongly continuous unitary one-parameter group. Stone's theorem therefore yields a unique self-adjoint generator H_ζ with

$$U_{e^t}^V = e^{itH_\zeta}.$$

Replacing e^t by λ gives the stated formula.

For a smooth compactly supported differential form ξ, differentiating the flow at $t = 0$ gives

$$\frac{d}{dt}\Big|_{t=0} U_{e^t}^V \xi = -\mathcal{L}_{v_{\text{scale}}}\xi.$$

Comparing with

$$\frac{d}{dt}\Big|_{t=0} e^{itH_\zeta}\xi = iH_\zeta\xi$$

yields $H_\zeta = -i\mathcal{L}_{v_{\text{scale}}}$ on the common core. □

Corollary 4.12 (Hermitian MBTT clause)**.** *The branch-specific MBTT clause*

$$\text{IsHermitian} : \Pi(u, v : V_\zeta)\ \text{Id}\big(\langle H_\zeta u, v\rangle, \langle u, H_\zeta v\rangle\big)$$

is discharged in $\mathcal{B}_{15}$ by geometry alone.

Proof. Self-adjointness from theorem 4.11 is exactly the required identity. □

5 The prime-trace theorem

5.1 Orbital trace formula

Definition 5.1 (Integrated scaling operators)**.** For $h \in \mathscr{S}$, define

$$U_V(h) := \int_0^\infty h(\lambda) U_\lambda^V \, d^\times\lambda, \qquad U_\Phi(h) := \int_0^\infty h(\lambda) U_\lambda^\Phi \, d^\times\lambda.$$

The operator $U_V(h)$ is the Hilbert-side realization used for spectral calculus, while $U_\Phi(h)$ is the dualizable test-side realization used for trace evaluation.

Lemma 5.2 (Regularized supertrace on the rigged carrier)**.** *For every $h \in \mathscr{S}$, the prime-trace value is evaluated through the regularized supertrace*

$$\text{Tr}_{\text{reg}}(U_\Phi(h)) := \text{Tr}_{\text{cat}}\big(U_\Phi(h)\big|_{\Phi_\zeta^{\text{even}}}\big) - \text{Tr}_{\text{cat}}\big(U_\Phi(h)\big|_{\Phi_\zeta^{\text{odd}}}\big).$$

It is not defined as a Hilbert-space Fredholm trace of $U_V(h)$.

Proof. By theorem 4.7, Φ_ζ is dualizable, so the categorical traces on its even and odd pieces are defined in the late cohesive shell. The Hilbert operator $U_V(h)$ records the same scaling dynamics on the completion, but no trace-class claim is made there. The alternating categorical trace on Φ_ζ is exactly the regularized cohomological trace exported by the public API. □

Definition 5.3 (Weil distribution)**.** Define the global Weil distribution on $\mathscr{S}$ by the local formula

$$W(h) := \sum_v \int'_{\mathbb{Q}_v^\times} \frac{h(u^{-1})}{|1-u|_v}\, d^\times u$$

Classically, the same distribution admits the alternate representation

$$W(h) = \widehat{h}(0) + \widehat{h}(1) - \sum_\rho \widehat{h}(\rho),$$

where ρ ranges over the non-trivial zeros of ζ, the prime in the local formula indicates the standard principal-value regularization at $u = 1$, and the sum over zeros is understood in the usual Weil-distribution sense rather than as an absolutely convergent series.

Theorem 5.4 (Prime-trace theorem)**.** *For every $h \in \mathscr{S}$,*

$$\mathrm{Tr}_{\mathrm{reg}}\big(U_\Phi(h)\big) = W(h).$$

Equivalently, the graded scaling action on the rigged datum $\Phi_\zeta \hookrightarrow V_\zeta$ realizes the Weil explicit distribution, with trace evaluation on Φ_ζ and spectral realization on V_ζ.

Proof. By theorem 4.7, the endomorphism $U_\Phi(h)$ acts on a dualizable test object, so its categorical trace is defined without any cutoff or singular-value estimate. The orbit classification of theorem 3.7 gives the finite-place contribution: every primitive closed orbit is indexed by a prime p, its m-fold iterate has length $m \log p$, and the corresponding categorical Lefschetz factor is the standard transverse term determined by the normal return map. Summing over all iterates yields exactly the finite-place term in Definition 5.3.

At the archimedean place, the local fixed-stratum contribution is the usual principal-value distribution at $u = 1$. Here no undeclared boundary projector is used: by theorem 4.5, the pole contribution is encoded by the concrete volume map I_{vol}, the trivial-zero jets are encoded by the concrete formal-neighborhood map J_∞, and the critical carrier is their exact common kernel. The complementary sector $V_{\mathrm{triv}} = V_\zeta^\perp$ is therefore the closed Hilbert span of explicit boundary data rather than a wished-for spectral summand. The common continuous scaling sector contributes equally to $\Phi_\zeta^{\mathrm{even}}$ and $\Phi_\zeta^{\mathrm{odd}}$ and therefore cancels in the supertrace. Because the trace is taken on Φ_ζ rather than on the noncompact Hilbert carrier, no commutator estimate with an auxiliary spatial cutoff enters. The total geometric trace is therefore the local sum

$$\sum_v \int'_{\mathbb{Q}_v^\times} \frac{h(u^{-1})}{|1-u|_v}\, d^\times u = W(h).$$

This is the PEN-internal form of the Connes–Meyer dictionary: Connes identifies the orbit-explicit-formula correspondence, Meyer realizes the same distribution cohomologically on a nuclear carrier, and the late PEN shell packages that realization as the categorical supertrace of Φ_ζ rather than as a trace-class statement on V_ζ [Connes, 1999, Meyer, 2005, Connes et al., 2009, Connes and Consani, 2023]. Hence

$$\mathrm{Tr}_{\mathrm{reg}}\big(U_\Phi(h)\big) = W(h).$$

□

Remark 5.5 (Why no Hilbert-space Lefschetz theorem is claimed)**.** One should not read $\mathrm{Tr}_{\mathrm{reg}}$ as the limit of $\mathrm{Tr}(P_\Lambda U_V(h) P_\Lambda)$ for spatial cutoffs on $L^2(X_\zeta)$. That classical promotion is exactly the analytic gap in the Connes–Meyer program: it would require controlling the continuous spectrum at infinity and the boundary commutators created by the cutoff. The present paper avoids that claim by placing the trace on the dualizable test object Φ_ζ and using V_ζ only for the self-adjoint generator and its spectral theorem.

5.2 Critical-line realization

Theorem 5.6 (Critical-line spectral representation)**.** *There exists a signed spectral measure σ_ζ on $\mathbb{R}$ such that, for every $h \in \mathscr{S}$,*

$$\mathrm{Tr}_{\mathrm{reg}}(U_\Phi(h)) = \widehat{h}(0) + \widehat{h}(1) - \int_{\mathbb{R}} \widehat{h}\Big(\tfrac{1}{2} + i\lambda\Big)\, d\sigma_\zeta(\lambda).$$

In particular, the spectral contribution coming from the self-adjoint generator H_ζ is supported on the line $\frac{1}{2} + i\mathbb{R}$.

Proof. By theorem 4.11, H_ζ is self-adjoint, so the spectral theorem gives a real spectral resolution $dE_\zeta(\lambda)$ on V_ζ. By theorem 4.7, the test-side action on Φ_ζ is densely embedded in V_ζ and is generated on the common core by the same Lie derivative whose closure is H_ζ. Thus $U_\Phi(h)$ and $U_V(h)$ encode the same scaling dynamics on the test and Hilbert sides. Applying functional calculus to H_ζ and pairing the resulting spectral resolution against the even-minus-odd multiplicity distribution induced from the dense grading defines a signed spectral measure σ_ζ with

$$\mathrm{Tr}_{\mathrm{reg}}(U_\Phi(h)) = \widehat{h}(0) + \widehat{h}(1) - \int_{\mathbb{R}} \widehat{h}\Big(\tfrac{1}{2} + i\lambda\Big)\, d\sigma_\zeta(\lambda).$$

The pole part of the boundary datum contributes $\widehat{h}(0) + \widehat{h}(1)$, while the jet-kernel condition defining Φ_ζ removes the archimedean trivial-zero sector from the critical trace. Meyer's cohomological realization places the non-trivial zero term in odd degree, hence with a minus sign.

This proves the claimed critical-line spectral representation. It should be read as a statement about the realized spectral side attached to H_ζ, not as a converse theorem reconstructing the full classical zero multiset from the distributional identity for W. Passing from equality of test-function functionals to isolation of individual zeros would require a separate Weil-criterion converse argument controlling the tail of the zero sum, and no such argument is supplied here. □

Corollary 5.7 (Remaining converse step for RH)**.** *If one additionally proves that the signed measure σ_ζ of theorem 5.6 recovers the full non-trivial zero distribution occurring in the alternate formula of Definition 5.3, then the Riemann Hypothesis follows. That converse identification is not proved in this manuscript.*

Proof. Under the stated additional identification, every spectral contribution to the zero term is supported on $\frac{1}{2} + i\mathbb{R}$ by theorem 5.6, so every non-trivial zero would lie on the critical line. □

6 AST optimality of the zeta operator

The regularized trace argument already settles the rigged prime-trace realization on the test side and the critical-line spectral representation on the Hilbert side. We now show that the same operator is also the unique PEN-optimal realization of the prime-trace API.

6.1 Hermitian and off-line shells

We compare only candidates that realize the same public interface of Definition 2.5. Shared public clauses are not charged in the branch comparison.

Definition 6.1 (Hermitian shell)**.** The Hermitian realization consists of the following two local clauses:
(H1) `H_zeta : End(V_zeta)`;
(H2) `IsHermitian`.
Thus

$$\kappa_{\mathrm{herm}} = 2.$$

Definition 6.2 (Minimal off-line shell)**.** A minimal off-line realization requires, in addition to operator formation, the following repair clauses:
(N1) Pair: explicit conjugate-pairing witnesses;
(N2) Reflect: explicit $s \leftrightarrow 1-s$ reflection witnesses;
(N3) ResCalc: non-self-adjoint resonance or holomorphic-calculus discipline;
(N4) Biorth: left/right spectral-coupling data for non-orthogonal modes.
Including operator formation, this gives

$$\kappa_{\mathrm{nh}} = 5.$$

Lemma 6.3 (Necessity of the repair clauses)**.** *Every off-line realization of the prime-trace API requires the four clause families of Definition 6.2.*

Proof. Without Pair, a non-real spectral term contributes a non-real regularized trace and cannot realize the real-valued distribution W. Without Reflect, the branch fails to implement the functional equation symmetry. Without ResCalc, the non-self-adjoint branch has no canonical calculus for x^{iH} or for the rigged integrated operators $U_\Phi(h)$ and $U_V(h)$. Without Biorth, the regularized trace of the non-normal spectral expansion is not computable from right eigenvectors alone. Each failure is independent; hence all four clause families are required. □

Theorem 6.4 (AST effort-gap theorem)**.** *For any off-line realization of the prime-trace API,*

$$\kappa_{\mathrm{nh}} - \kappa_{\mathrm{herm}} \geq 3.$$

In the minimal comparison of Definitions 6.1 and 6.2, the gap is exactly

$$\Delta\kappa = 5 - 2 = 3.$$

Proof. The Hermitian shell needs only operator formation and self-adjointness. By theorem 6.3, every off-line shell requires, beyond operator formation, at least the three additional clause families Pair, Reflect, and ResCalc; the explicit comparison also requires Biorth. Therefore the effort gap is at least three, and equals three in the minimal branch count above. □

6.2 Novelty monotonicity at fixed public API

Theorem 6.5 (No extra public novelty)**.** *Let X_{herm} and X_{nh} be Hermitian and off-line realizations of the same prime-trace API. Then*

$$\nu(X_{\mathrm{nh}}) \leq \nu(X_{\mathrm{herm}}).$$

Proof. The two candidates export the same public interface: the rigged pair $\Phi_\zeta \hookrightarrow V_\zeta$, the scaling actions on both carriers, the integrated operators $U_\Phi(h)$ and $U_V(h)$, the regularized trace functional $\mathrm{Tr}_{\mathrm{reg}}$, and the evaluation law $\Theta(h)$. The off-line clauses Pair, Reflect, ResCalc, and Biorth are internal repair witnesses needed only to make the non-Hermitian branch type-check and compute. They introduce no new public carrier, eliminator, or interaction principle. Therefore they contribute no additional public schema families beyond the fixed API, whence the claimed inequality. □

Corollary 6.6 (PEN dominance of the Hermitian shell)**.** *Among all realizations of the prime-trace API, PEN selects the Hermitian shell.*

Proof. By theorem 6.4, the non-Hermitian branch has strictly larger effort. By theorem 6.5, it has no greater public novelty. Hence its efficiency ratio $\rho = \nu/\kappa$ is strictly smaller. PEN therefore selects the Hermitian branch. □

7 Final synthesis and the remaining converse step

We can now state the combined analytic and syntactic conclusion.

Theorem 7.1 (PEN-universe prime-trace synthesis)**.** *In the step-14/15 PEN universe:*

(i) the synthetic idele-class carrier X_ζ exists as a derived object of the selected library;

(ii) the scaling action of $\mathbb{R}_{>0}$ on X_ζ yields a self-adjoint prime-trace operator H_ζ on the critical graded Hilbert space V_ζ and a compatible trace action on the dense test module Φ_ζ;

(iii) the regularized categorical supertrace of the integrated scaling action on Φ_ζ realizes the Weil explicit distribution;

(iv) the same regularized trace admits a critical-line spectral representation by a signed measure σ_ζ on $\mathbb{R}$ determined by the self-adjoint generator H_ζ.

Moreover the realizing shell is the unique PEN-optimal branch of the public prime-trace API.

Proof. Items (i)–(iii) are theorems 3.4, 4.11 and 5.4. Item (iv) is theorem 5.6. The uniqueness statement is theorem 6.6. □

Remark 7.2 (What is not claimed)**.** Theorem 7.1 does not identify σ_ζ with the full classical zero distribution appearing in the alternate representation of W. Establishing that converse implication would require a separate Weil-criterion argument beyond the present manuscript; compare theorem 5.7.

8 Mechanization path against the public artifact

The current public repository already separates the formal and computational pillars of PEN: Cubical Agda for theorem-level contracts and Haskell for typed candidate generation and scoring [Lande, 2026f,a]. The rigged prime-trace construction fits that split naturally.

On the Agda side, the required formal layers are:

1. the late-universe record exposing the step-14/15 shell interfaces;
2. the MBTT clause-count comparison for theorem 6.4;
3. the proof that off-line repair clauses add no public schema families;
4. the abstract rigged-supertrace and critical-line representation proof of theorem 5.6;
5. the final theorem record stating theorem 7.1 together with a separate interface for any future Weil-converse strengthening.

On the Haskell side, one adds a candidate shell matching the appendix below and verifies that the strict scorer assigns the local clause count 2 to the Hermitian shell and at least 5 to the off-line shell. The scoring comparison is finite and objective; it does not depend on any numerical approximation of zeta zeros.

9 Conclusion

The zeta prime-trace problem becomes rigid once it is asked in the correct late-stage PEN language. The right object is not a list of zeros and not a raw trace of a fixed operator. It is a rigged prime-trace API living in the metric/Hilbert and temporal-cohesive environment selected by the Fibonacci growth law. Inside that environment the geometry of the idele-class scaling flow yields a self-adjoint generator on the Hilbert completion V_ζ and a dualizable codiscrete test module Φ_ζ on which the adelic trace formula is evaluated. The continuous scaling sector cancels in the categorical even-minus-odd trace on Φ_ζ, while the self-adjoint spectral parameter on V_ζ yields a spectral representation supported on $\Re(s) = 1/2$.

What this paper does not claim is the separate converse step needed to identify that realized spectral measure with the full classical zero distribution. That remains an additional theorem, not a consequence of the distributional trace identity alone.

The same operator is also the unique PEN-optimal shell: any off-line branch must add repair clauses for pairing, reflection, and non-self-adjoint resonance calculus while exporting no additional public novelty. Thus the Hermitian critical-line spectral package is not only geometrically singled out by the rigged prime-trace realization; it is also the only branch that survives the structural optimization rule governing the PEN universe.

A MBTT pseudo-code for the winning shell

Listing 1: The Hermitian zeta shell in the late PEN universe.

```
-- ================================================================
-- PEN candidate shell: ZetaHermitian
-- ================================================================

-- 1. Geometric provenance (derived, hence uncharged)
Z_geom  : U = Omega(S1)
Z_hat   : U = StoneCompletion(Z_geom)
A_fin   : U = SyntheticFiniteAdeles(Z_hat)
X_zeta  : U = Quotient(Prod(R_derived, A_fin), Units(Q_geom))

-- 2. Rigged Hilbert-functional lift
g_zeta  : Metric  = ScaleInvariantMetric(X_zeta)
V_full  : Hilbert = App deRhamL2 (X_zeta, g_zeta)
Phi_full : CohModule = App Sharp (SmoothForms X_zeta)
x_arch  : Point(ArchCompactify X_zeta) = ArchBoundaryPoint(X_zeta)
Vol_int : LinearMap(Phi_full, R)
Vol_int phi = IntegrateTop(phi, g_zeta)

D_inf   : CohSpace = FormalNeighborhood(x_arch)
Jet_eval : LinearMap(Phi_full, SmoothForms D_inf)
Jet_eval phi = Restrict(phi, D_inf)

Phi_zeta : CohModule = Intersect(Kernel(Vol_int), Kernel(Jet_eval))
V_zeta   : Hilbert = HilbertClosure(ImageToL2(Phi_zeta), V_full)
V_triv   : ClosedSubspace(V_full) = OrthogonalComplement(V_zeta)
P_zeta   : Projector = OrthogonalProjector(V_full, V_zeta)
Phi_even : CohModule = EvenForms(Phi_zeta)
Phi_odd  : CohModule = OddForms(Phi_zeta)
Embed    : DenseMap(Phi_zeta, V_zeta)

-- 3. Temporal/cohesive flow
U_scale_V : Pi(x : R_gt_0) -> End(V_zeta)
U_scale_V x = PullbackScaleAction(x, X_zeta)

U_scale_Phi : Pi(x : R_gt_0) -> End(Phi_zeta)
U_scale_Phi x = Restrict(App U_scale_V x, Phi_zeta)

Inv_Phi : IsInvariant(U_scale_Phi, Phi_zeta)
Inv_Phi = pf_IntersectInvariant(pf_VolKernelInvariant, pf_JetKernelInvariant)

H_zeta  : End(V_zeta)
H_zeta  = StoneGenerator(U_scale_V)

U_int_V   : Pi(h : TestFn) -> End(V_zeta)
U_int_V h = IntegratedScaleAction(U_scale_V, h)

U_int_Phi   : Pi(h : TestFn) -> End(Phi_zeta)
```

```
U_int_Phi h = IntegratedScaleAction(U_scale_Phi, h)

TraceReg : Pi(h : TestFn) -> R
TraceReg h = Sub(CatTrace(Restrict(App U_int_Phi h, Phi_even)),
                 CatTrace(Restrict(App U_int_Phi h, Phi_odd)))

Theta   : Pi(h : TestFn) -> Id(TraceReg h, Weil(h))
Theta h = pf_CohomologicalLefschetzTrace(Phi_zeta, h)

-- 4. Winning Hermitian clause
IsHermitian : Pi(u : V_zeta) (v : V_zeta)
            -> Id(Inner(App H_zeta u, v), Inner(u, App H_zeta v))
IsHermitian = pf_StoneGeneratorSelfAdjoint(U_scale_V)
```

B Minimal off-line repair shell

Listing 2: Minimal off-line repair clauses.

```
-- Shared formation clause omitted from comparison.

Pair      : Pi(r : Spec(H_zeta)) -> Sigma(r_star : Spec(H_zeta))
            (Id r_star (Conj r))

Reflect   : Pi(r : Spec(H_zeta)) -> Sigma(r_ref : Spec(H_zeta))
            (Id r_ref (Subtract 1 r))

ResCalc   : NonSelfAdjointFunctionalCalculus(H_zeta)

Biorth    : LeftRightCoupling(H_zeta)
```

References

Enrico Bombieri. The riemann hypothesis: Official problem description, 2000. Clay Mathematics Institute.

Felix Cherubini, Thierry Coquand, Freek Geerligs, and Hugo Moeneclaey. A foundation for synthetic stone duality. In *TYPES 2024*. Leibniz International Proceedings in Informatics, 2025.

Alain Connes. Trace formula in noncommutative geometry and the zeros of the riemann zeta function. *Selecta Mathematica*, 5(1):29–106, 1999.

Alain Connes and Caterina Consani. Weil positivity and trace formula, the archimedean place. *Journal of Number Theory*, 248:1–61, 2023.

Alain Connes, Caterina Consani, and Matilde Marcolli. The weil proof and the geometry of the adeles class space. In *Algebra, Arithmetic, and Geometry*, pages 339–405. Birkh"auser, 2009.

Halvor Lande. Pen repository architecture. Repository documentation, 2026a. URL https://github.com/efficient-novelty/mechanization.

Halvor Lande. Automated theory synthesis via typed structural optimization, 2026b. Manuscript.

Halvor Lande. Coherence depth and fibonacci scaling in intensional type theory, 2026c. Manuscript.

Halvor Lande. Synthetic framework abstraction and the temporal-cohesive terminal fixed point, 2026d. Manuscript.

Halvor Lande. A generative calculus for type-theoretic complexity and the geometric core, 2026e. Manuscript.

Halvor Lande. Pen theory: Mechanized from first principles. GitHub repository, 2026f. URL https://github.com/efficient-novelty/mechanization.

Daniel Licata and Michael Shulman. Calculating the fundamental group of the circle in homotopy type theory. *Logical Methods in Computer Science*, 13(1), 2017.

Ralf Meyer. A spectral interpretation for the zeros of the riemann zeta function. *Mathematisches Institut, Georg-August-Universit"at G"ottingen: Seminars Winter Term 2004/2005*, pages 117–137, 2005.

E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, 2 edition, 1986.